

Trade-offs, Comparative Advantage, and the Market System

ECON 101 - Macroeconomics

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- 2.1 Production Possibilities Frontiers and Opportunity Costs
- 2.2 Comparative Advantage and Trade
- 2.3 The Market System

Scarcity and Trade-offs

- People continually face decisions about how best to use their **scarce** resources.
- **Scarcity**: a situation in which unlimited wants exceed the limited resources available to fulfill those wants.
- Scarcity requires **trade-offs**: to get more of one thing, society must accept less of another.
- Economics gives us tools to help make these trade-offs wisely.

Example: Tesla and Trade-offs

- Tesla has scarce workers and machinery.
- If Tesla wants to produce more Model 3 sedans, those resources are not available to produce Cybertruck pickups (and vice versa).
- This illustrates the central macro idea: even a high-tech firm cannot produce unlimited quantities of everything at once.

2.1 Learning Objective

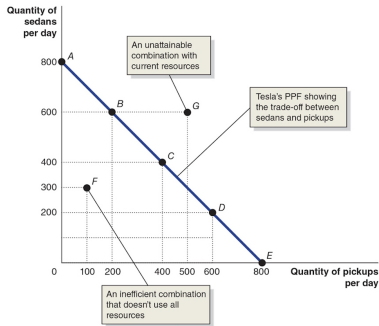
Goal

Use a production possibilities frontier to analyze opportunity costs and trade-offs.

- A **production possibilities frontier (PPF)** is a curve showing the maximum attainable combinations of two goods that can be produced with available resources and current technology.
- The PPF is a **positive** tool: it shows what *is* possible, not what *should* be produced.

Figure 2.1 – Tesla's Production Possibilities Frontier (1 of 2)

Tesla's Production Possibilities at Its Plant		
Choice	Quantity of Model 3 Sedans Produced	Quantity of Cybertruck Pickups Produced
A	800	0
B	600	200
C	400	400
D	200	600
E	0	800
F	300	100
G	600	500

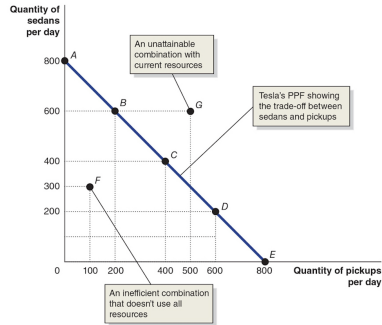


Interpreting the PPF: Tesla Example

- Tesla can produce combinations of Model 3 sedans and Cybertruck pickups.
- **Points on** the PPF: attainable and **productive efficient**.
- **Points below** the PPF: attainable but **inefficient** (resources underused).
- **Points above** the PPF: currently unattainable with existing resources and technology.

Figure 2.1 – Tesla's PPF and Opportunity Cost (2 of 2)

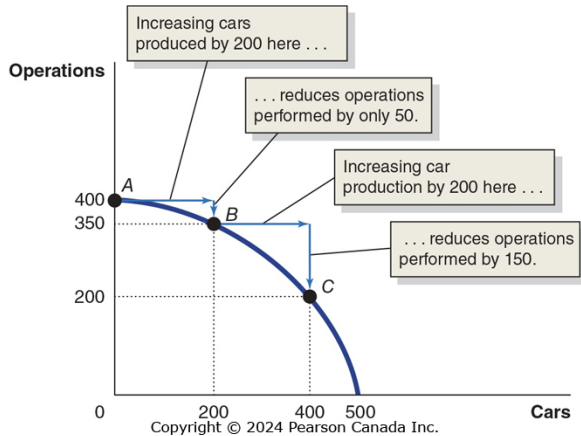
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Opportunity Cost on the PPF

- Suppose moving from point A to point B:
 - Tesla produces 200 more Cybertruck pickups.
 - To do so, it must produce 200 fewer Model 3 sedans.
- The **opportunity cost** of producing 200 more Cybertrucks is the 200 Model 3 sedans given up.
- **Opportunity cost:** the highest-valued alternative that must be given up to engage in an activity.

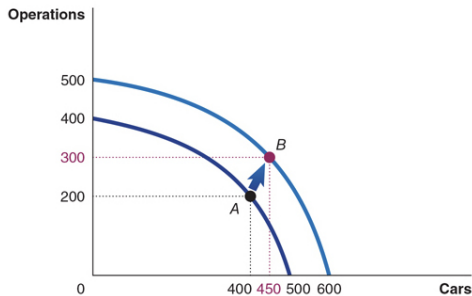
Figure 2.2 – Increasing Marginal Opportunity Costs



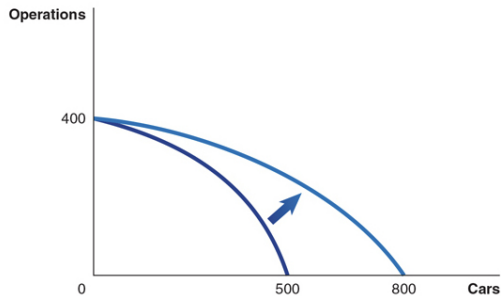
Why Opportunity Costs Often Increase

- Some resources are better suited to one task than another.
- The first resources to be switched are the ones best suited to switching.
- As more resources are devoted to an activity, the extra output from additional resources tends to fall.
- This gives the PPF its bowed-out shape and leads to **increasing marginal opportunity cost**.

Figure 2.3 – Economic Growth



(a) Shifting out the PPF

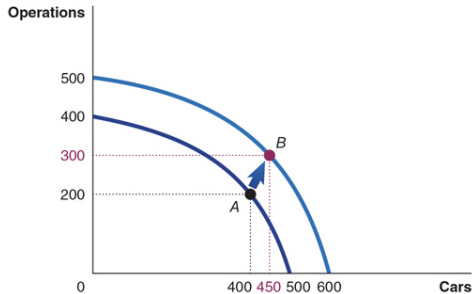


(b) Technological change in the automotive industry

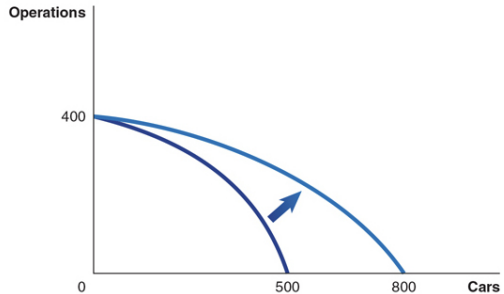
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- As more economic resources become available, the economy can move from point A to point B, producing more of both goods.
- Shifts in the PPF represent **economic growth**.
- **Economic growth**: the ability of the economy to increase the production of goods and services.
- Macro link: long-run growth allows higher living standards without giving up as much of other goods.

Figure 2.3 – Technological Change in One Sector



(a) Shifting out the PPF



(b) Technological change in the automotive industry

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Technological Change in One Industry

- Suppose technology improves in the automobile industry only.
- The maximum output of automobiles increases, but the production possibility of the other good (e.g., tanks) is unchanged.
- Previously unattainable combinations become attainable.
- This captures sector-specific technological progress common in macroeconomics (e.g., ICT revolution).

2.2 Learning Objective

Goal

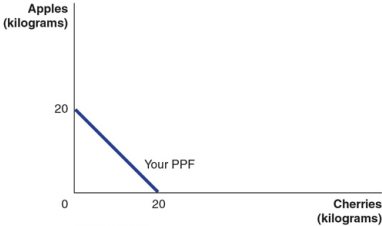
Understand comparative advantage and explain how it serves as the basis for trade.

Two-Person Production Example: You and Your Neighbour

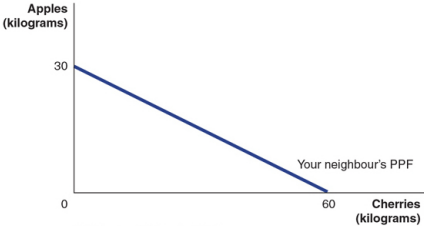
- You and your neighbour each have limited time to pick apples and/or cherries.
- The table (in the slides) shows:
 - If you spend all your time picking cherries, you can pick 20 kg of cherries.
 - If you spend all your time picking apples, you can pick 20 kg of apples.
 - Your neighbour can pick 60 kg of cherries or 30 kg of apples if she specializes.
- We use these numbers to illustrate absolute and comparative advantage.

Figure 2.4 – Production Possibilities for You and Your Neighbour, Without Trade

	You		Your Neighbour	
	Apples (kg)	Cherries (kg)	Apples (kg)	Cherries (kg)
Pick nothing but apples	20	0	30	0
Pick nothing but cherries	0	20	0	60



(a) Your PPF



(b) Your neighbour's PPF

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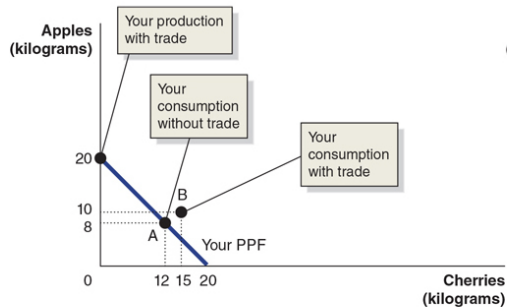
Autarky (No Trade) Outcomes

- Without trade, each person consumes only what they produce.
- **You:** pick and consume 8 kg of apples and 12 kg of cherries per week (point A).
- **Neighbour:** picks and consumes 9 kg of apples and 42 kg of cherries per week (point C).
- Both are on their own PPFs but may not be at the best possible consumption points if trade is allowed.

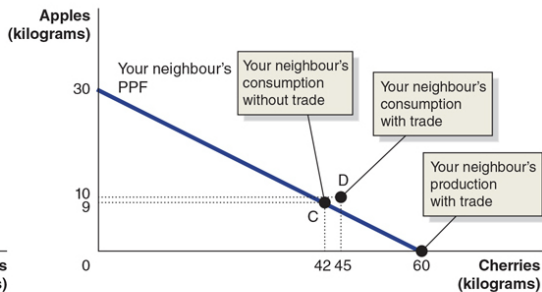
Specialization and Gains from Trade

- **Trade:** the act of buying and selling.
- What if you and your neighbour decide to specialize and trade?
- Your neighbour appears better at picking both apples and cherries, but:
 - You can still both benefit from trade by specializing in what you are **relatively** good at.

Figure 2.5 – Gains from Trade (1 of 2)



(a) Your production and consumption with trade



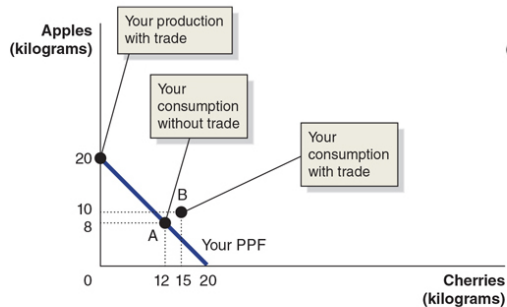
(b) Your neighbour's production and consumption with trade

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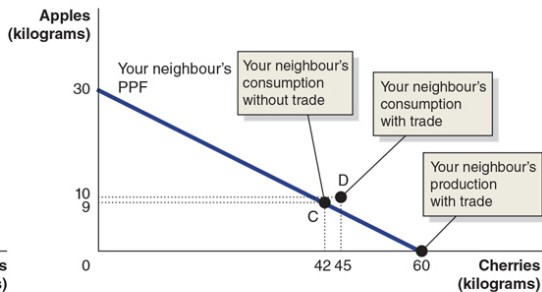
Numerical Trade Example

- If you specialize in picking apples, you can pick 20 kg.
- If your neighbour specializes in picking cherries, she can pick 60 kg.
- Suppose you trade 10 kg of your apples for 15 kg of your neighbour's cherries:
 - **You** consume 10 kg of apples and 15 kg of cherries (point B).
 - **Neighbour** consumes 10 kg of apples and 45 kg of cherries (point D).
- Both are better off than in autarky.

Figure 2.5 – Gains from Trade (2 of 2)



(a) Your production and consumption with trade



(b) Your neighbour's production and consumption with trade

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Table 2.1 – Summary of the Gains From Trade

Table 2.1 A Summary of the Gains from Trade

	You		Your Neighbour	
	Apples (kg)	Cherries (kg)	Apples (kg)	Cherries (kg)
Production and consumption <i>without</i> trade	8	12	9	42
Production <i>with</i> trade	20	0	0	60
Consumption <i>with</i> trade	10	15	10	45
Increased consumption (gains from trade)	2	3	1	3

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Table 2.1 – Summary of the Gains From Trade

- Both you and your neighbour are able to consume more with trade than without trade.
- Key points to emphasize to students:
 - Trade allows consumption points outside each individual's own PPF.
 - Even if one person is better at producing *both* goods, both can gain.

Explaining the Gains from Trade

- Your neighbour has an **absolute advantage** in both cherry and apple picking.
- You have a **comparative advantage** in picking apples.
- **Absolute advantage:** ability to produce more of a good or service than competitors using the same amount of resources.
- **Comparative advantage:** ability to produce a good or service at a lower opportunity cost than competitors.

Table 2.2 – Opportunity Costs of Picking Apples and Cherries

Table 2.2 Opportunity Costs of Picking Apples and Cherries

	Opportunity Cost of Apples	Opportunity Cost of Cherries
You	1 kg of cherries	1 kg of apples
Your neighbour	2 kg of cherries	0.5 kg of apples

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Table 2.2 – Opportunity Costs of Picking Apples and Cherries

- This table compares the opportunity cost of apples vs. cherries for you and your neighbour.
- The key lesson:
 - The basis for trade is **comparative advantage**, not absolute advantage.
 - Individuals, firms, and countries are better off if they specialize in producing goods and services for which they have a comparative advantage and obtain the other goods and services they need by trading.

2.3 Learning Objective

Goal

Explain the basic idea of how a market system works.

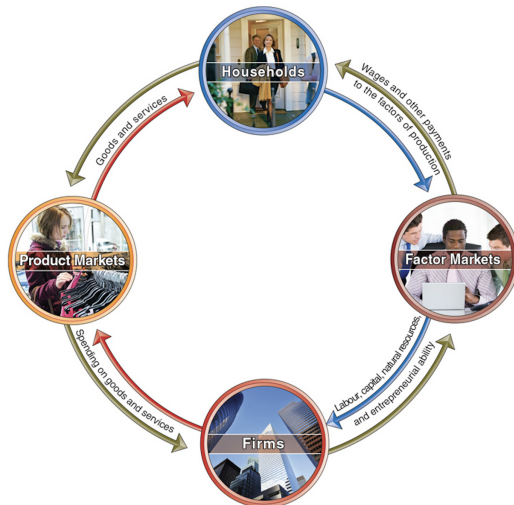
- A **market** is a group of buyers and sellers of a good or service, and the institution or arrangement by which they come together to trade.
- Two key groups participate in the modern economy:
 - **Households**
 - **Firms**

- **Households** consist of individuals who provide the **factors of production**:
 - labour, capital, natural resources, and entrepreneurial ability.
- Households receive payments for these factors by selling them to firms in **factor markets**.
- **Firms** supply goods and services to **product markets**; households buy these products from the firms.

The Four Factors of Production

- **Labour:** all types of work, from part-time jobs at McDonalds to CEOs of large corporations.
- **Capital:** physical capital such as computers, machines, and buildings used to produce other goods.
- **Natural resources:** land, water, oil, iron ore, and other raw materials (“gifts of nature”).
- **Entrepreneur:** someone who operates a business.
- **Entrepreneurial ability:** ability to bring together the other factors of production to produce and sell goods and services successfully.

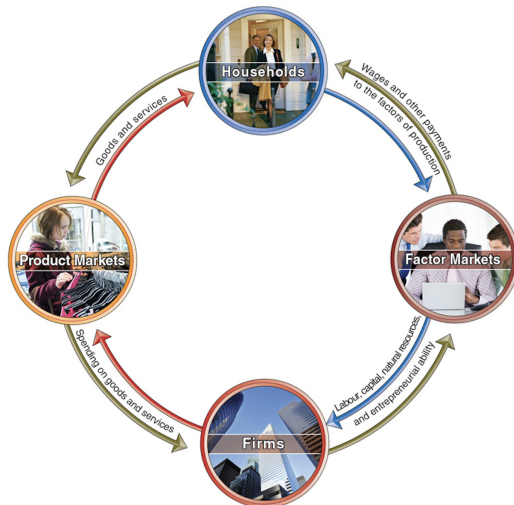
Figure 2.6 – The Circular-Flow Diagram (1 of 2)



Circular-Flow Diagram: Basic Idea

- The **circular-flow diagram** is a model that illustrates how participants in markets are linked.
- **Real flows:**
 - Households provide factors of production to firms.
 - Firms provide goods and services to households.
- **Money flows:**
 - Firms pay money to households for the factors of production.
 - Households pay money to firms for goods and services.

Figure 2.6 – The Circular-Flow Diagram (2 of 2)



Limits of the Simple Circular-Flow Model

- Like all economic models, the circular-flow diagram is a **simplified** version of reality.
- The basic version excludes:
 - Government
 - Financial system (banks, credit markets)
 - Foreign buyers and sellers of goods and services
- We add these sectors later in the course to build full macro models.

- A **free market** has few government restrictions on:
 - how goods or services can be produced or sold, or
 - how factors of production can be employed.
- Countries that come closest to this benchmark have typically been more successful than centrally planned economies in raising living standards.
- Classic idea: Adam Smith's "**invisible hand**" – individuals acting in their own self-interest collectively satisfy the wants of consumers.

Things to Remember

- The production possibilities frontier should **never bow inward**. Why?
 - Because opportunity cost typically increases as you move along the PPF.
- The PPF tells us what **can** be produced, not what **should** be produced.
- Just because someone is better or worse at *everything* does not mean trade with them cannot be beneficial.
- The basis for trade is **comparative advantage**, not absolute advantage.
- Free markets raise the standard of living, but that does not mean there is no role for governments:
 - Governments must provide a sound legal environment for the market system to succeed.

- PPFs illustrate trade-offs, opportunity cost, economic growth, and technological change.
- Comparative advantage explains why individuals and countries gain from specialization and trade, even when one side has an absolute advantage in everything.
- The market system coordinates decisions of households and firms through product and factor markets.
- The circular-flow diagram provides a simple macro picture of how goods, services, and money move in the economy.
- Free markets and well-functioning institutions are central to long-run improvements in living standards.

Thank you!